

measurable = Physical quantities

Fundamental Quantities

- 1) mass - kg
- 2) Length - m
- 3) Time - s
- 4) Temperature - K
- 5) mole - mol
- 6) Luminous Intensity - cd
- 7) Current - Ampere

Scalar Quantities

→ Quantities which have only magnitude.

Eg:- Dist, speed, Temp, mass, pressure

Vector Quantities

→ They have both magnitude and direction.

Eg:- Disp, velo, acc, Force.

S.I unit REFERENCE

Large Positive Numbers

K - Kilo - 10^3

M - Mega - 10^6

G - Giga - 10^9

T - Tera - 10^{12}

P - Peta - 10^{15}

Small Negative Powers

d - deci - 10^{-2}

c - centi - 10^{-2}

m - milli - 10^{-3}

μ - micro - 10^{-6}

n - nano - 10^{-9}

p - pico - 10^{-12}

f - femto - 10^{-15}

special unit:- Not a prefix but used everywhere

A - Armstrong - $10^{-10}m$

Dimensions: The dimensions of a quantity are the powers to which MLT must be raised to represent it.

Notation:

$[M], [L], [T]$

$$\text{Speed} = \frac{\text{Dist}}{\text{Time}} = \frac{\text{Length}}{\text{Time}} = \frac{[L]}{[T]} = [LT^{-1}] = [m^0 L T^{-1}]$$

$$\text{Area} = \text{Length} \times \text{Length} = [L^2] = [m^0 L^2 T^0]$$

$$\text{Volume} = L \times L \times L = [L^3] = [m^0 L^3 T^0]$$

$$\text{Acc}^n = \frac{\text{Velocity}}{\text{Time}} = \frac{[LT^{-1}]}{[T]} = [LT^{-2}]$$

$$\text{Force} = ma = [m] [LT^{-2}] = [mLT^{-2}]$$

$$\text{Work} = F \cdot S = [mLT^{-2}] [LT^0] = [mL^2 T^{-2}]$$

★ Core mechanics Dimensions

$$1) \text{Work} = \text{Force} \times \text{Dist} = [m] [LT^{-2}] \times [L] = [mL^2 T^{-2}]$$

$$2) \text{Energy} = \frac{1}{2} mv^2 = \frac{1}{2} [m] [LT^{-1}]^2 = [m] [L^2 T^{-2}] = [mL^2 T^{-2}]$$

$$3) \text{Torque} = \text{Force} \times \text{Dist} = [mL^2 T^{-2}]$$

$$4) \text{Power} = \frac{\text{Work}}{\text{Time}} = \frac{[mL^2 T^{-2}]}{[T]} = [mL^2 T^{-3}]$$

$$5) \text{Impulse} = \text{Force} \times \text{Time} = [mLT^{-1}]$$

$$6) \text{Momentum} = \text{mass} \times \text{velo} = [mLT^{-1}]$$

$$7) \text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{[mLT^{-2}]}{[L^2]} = [mL^{-1} T^{-2}]$$

Harshith

S.I unit REFERENCE

Harshith

Large Positive Numbers

K - Kilo - 10^3

M - Mega - 10^6

G - Giga - 10^9

T - Tera - 10^{12}

P - Peta - 10^{15}

Small Negative Powers

d - deci - 10^{-1}

c - centi - 10^{-2}

m - milli - 10^{-3}

μ - micro - 10^{-6}

n - nano - 10^{-9}

p - pico - 10^{-12}

f - femto - 10^{-15}

Special unit:- Not a prefix but used everywhere

A - Armstrong - $10^{-10}m$

Dimensions: The dimensions of a quantity are the powers to which MLT must be raised to represent it.

★ Notation:

Harshith

$$[m], [L], [T]$$

$$\text{speed} = \frac{\text{Dist}}{\text{Time}} = \frac{\text{length}}{\text{Time}} = \frac{[L]}{[T]} = [LT^{-1}] = [m^0 L T^{-1}]$$

$$\text{Area} = \text{Length} \times \text{Length} = [L^2] = [m^0 L^2 T^0]$$

$$\text{volume} = L \times L \times L = [L^3] = [m^0 L^3 T^0]$$

$$\text{Acc}^n = \frac{\text{velocity}}{\text{Time}} = \frac{[LT^{-1}]}{[T]} = [LT^{-2}]$$

$$\text{Force} = ma = [m] [LT^{-2}] = [mLT^{-2}]$$

$$\text{work} = F \cdot s = [mLT^{-2}] [LT^0] = [mL^2 T^{-2}]$$

★ Core mechanics Dimensions

$$1) \text{ work} = \text{Force} \times \text{Dist} = [m] [LT^{-2}] \times [L] = [mL^2 T^{-2}]$$

$$2) \text{ Energy} = \frac{1}{2} mv^2 = \frac{1}{2} [m] [LT^{-1}]^2 = [m] [L^2 T^{-2}] = [mL^2 T^{-2}]$$

$$3) \text{ Torque} = \text{Force} \times \text{Dist} = [mL^2 T^{-2}]$$

$$4) \text{ Power} = \frac{\text{work}}{\text{Time}} = \frac{[mL^2 T^{-2}]}{[T]} = [mL^2 T^{-3}]$$

$$5) \text{ Impulse} = \text{Force} \times \text{Time} = [mLT^{-1}]$$

$$6) \text{ Momentum} = \text{mass} \times \text{velo} = [mLT^{-1}]$$

$$7) \text{ Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{[mLT^{-2}]}{[L^2]} = [mL^{-1} T^{-2}]$$

$$8) \text{ Density} = \frac{\text{mass}}{\text{Volume}} = \frac{[M]}{[L]^3} = [ML^{-3}]$$

Harshith

$$9) \text{ Frequency} = \frac{1}{T} = [T^{-1}]$$

$$10) \text{ Surface tension} = \frac{\text{Force}}{\text{Length}} = \frac{[MLT^{-2}]}{[L]} = [MT^{-2}]$$

*) Tricky Dimensions for JEE :-

$$1) \text{ Gravitational constant (G)} : G = \frac{F r^2}{m_1 m_2} = \frac{[M^1 L^1 T^{-2}] [L^2]}{[M] [M]} \\ = [M^{-1} L^3 T^{-2}]$$

2) Planck's constant :-

$$E = hf \rightarrow h = \frac{E}{f} = \frac{[ML^2 T^{-2}]}{[T^{-1}]} = [ML^2 T^{-1}]$$

3) Boltzmann Constant

$$E = kT \Rightarrow k = \frac{E}{T} = \frac{[ML^2 T^{-2}]}{[K]} = [ML^2 T^{-2} K^{-1}]$$

$\downarrow$
Energy
 $\downarrow$
Temperature

$$\text{Angular momentum} = mvr$$

Dimensions Less Quantities:

$$1) \text{ strain} = \frac{\text{Change in length}}{\text{original length}} = \frac{[L]}{[L]} = [M^0 L^0 T^0]$$

$$2) \text{ Refractive Index} = \frac{c}{v} = \frac{[L T^{-1}]}{[L T^{-1}]} = [M^0 L^0 T^0]$$

3) Relative Density = $\frac{\text{Density of sub}}{\text{Density of H}_2\text{O}} = [M^0 L^0 T^0]$ Harshith

4) Angle = $\frac{\text{Arc length}}{\text{Radius}} = [M^0 L^0 T^0]$

Same Dimensions

$[M^1 L^2 T^{-2}] = \text{Work, KE, PE, Torque, Heat}$

$[M^1 L^1 T^{-1}] = \text{momentum, Impulse}$

$[M^1 L^{-1} T^{-2}] = \text{Pressure, stress, Elastic modulus}$

$[M^1 L^2 T^{-1}] = \text{Planck's constant (h), Angular momentum}$

★ The Principle of Homogeneity :-

$[L] + [L] = [L] \text{ valid } \checkmark$

$[L] + [T] = \text{Meaningless } \times$

Def:- In any valid Physical equation, every term separated by a (+) or (-) sign and both sides of the equation must have the same dimensions.

Ex:-

1) $v = u + at$

LHS = $v = [LT^{-1}]$

RHS = $u = [LT^{-1}]$

at $[LT^{-2}][T]$

$= [LT^{-1}]$

$$2) S = ut + \frac{1}{2}at^2$$

Harshith

$$\text{LHS: } S = [L]$$

$$\text{RHS: } ut = [LT^{-1}][T+1] \\ = [L]$$

$$at^2 = [LT^{-2}][T+2] \\ = [L]$$

$$\text{Ex: (3) } v^2 = u^2 + 2as$$

$$\text{LHS: } v^2 = [L^2T^{-2}]$$

$$\text{RHS: } u^2 = [L^2T^{-2}]$$

$$2as = [LT^{-2}][L]$$

$$= [L^2T^{-2}]$$

$$\text{Ex: (6) } F = \frac{mv^2}{r}$$

$$\text{LHS: } F = [mLT^{-2}]$$

$$\text{RHS: } \frac{mv^2}{r} = \frac{[mL^2T^{-2}]}{[L]}$$

$$= [mLT^{-2}]$$

$$\text{Ex: (4) } v = u + \frac{1}{2}at^2$$

$$\text{LHS: } v = [LT^{-1}]$$

$$\text{RHS: } u = [LT^{-1}]$$

$$\frac{1}{2}at^2 = [LT^{-2}][T^2]$$

$$= [L]$$

Invalid

$$\text{Ex: (5) } v = u + 2at$$

$$\text{LHS: } v = [LT^{-1}]$$

$$\text{RHS: } u = [LT^{-1}]$$

$$2at = [LT^{-2}][T]$$

$$= [LT^{-1}]$$

Dimensional Analysis can prove an equation is wrong. (If terms don't match). It can NEVER prove an equation is right because it cannot detect errors hidden in dimensionless numbers.

Terms don't match $\rightarrow$ eqn is definitely wrong.

Terms match $\rightarrow$ eqn might be correct (not guarantee)

Ex: ⑦ $T = 2\pi\sqrt{\frac{L}{g}} \rightarrow$ Time period of pendulum

$$\text{LHS} = T = [T]$$

$$\text{RHS} = \frac{L = [L]}{g = [LT^{-2}]} = [\sqrt{T^2}] = [T]$$

* Conversion of units using dimensions:-

$$n_1 U_1 = n_2 U_2$$

$$n_2 = n_1 \left[\frac{m_1}{m_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

① Energy

SI unit

Joule (J)

CGS

Ergs

$$\rightarrow [ML^2T^{-2}]$$

SI units: Kg, ms

CGS units: gm, cm, s

$$n_2 = n_1 \left[\frac{m_1}{m_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

$$n_2 = 1 \left[\frac{1 \text{ kg}}{1 \text{ kg}} \right]^1 \left[\frac{1 \text{ m}}{1 \text{ cm}} \right]^2 \left[\frac{1 \text{ s}}{1 \text{ s}} \right]^{-2}$$

$$n_2 = \left[\frac{1000 \text{ gm}}{1 \text{ gm}} \right]^1 \left[\frac{100 \text{ cm}}{1 \text{ cm}} \right]^2 \left[\frac{1 \text{ s}}{1 \text{ s}} \right]^{-2}$$

$$n_2 = 10^3 \times (10^2)^2 \times 1^{-2}$$

$$= 10^3 \times 10^4 \times 1^{-2}$$

$$1 \text{ Joule} = 10^7 \text{ ergs}$$

② Force

SI
↓
Newton
(N)

Cgs
↓
Dyne

$$\rightarrow [MLT^{-2}]$$

$$a=1, b=1, c=-2$$

$$n_2 = n_1 \left[\frac{m_1}{m_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$$

$$n_2 = 1 \left[\frac{1000 \text{ gm}}{1 \text{ gm}} \right]^1 \left[\frac{100 \text{ cm}}{1 \text{ cm}} \right]^1 \left[\frac{1 \text{ s}}{1 \text{ s}} \right]^{-2}$$

$$= 10^3 \times 10^2 \times 1$$

$$n_2 = 10^5$$

$$1 \text{ N} = 10^5 \text{ Dyne}$$

* Deriving Formula:-

Harshith

Power Law

$$\Theta = KA^xB^yC^z$$

Θ = Quantity which we want to find out

K = Dimensionless quantity

$A, B, C \Rightarrow$ Quantities Θ is depend on

$x, y, z \rightarrow$ unknown powers

1) $T = Km^n L^y g^z$

LHS : $[T]$

RHS : $m = [m], L[L]$

$g = [LT^{-2}]$

$$[T] = [m]^x [L]^y [LT^{-2}]^z$$

$$[T] = [m]^x [L]^{1+y} [T]^{-2z}$$

$$0 = x \rightarrow ①$$

$$0 = y + 2 \rightarrow ②$$

$$0 = -2z \rightarrow ③$$

$$-2y = 1 \Rightarrow z = -\frac{1}{2}$$

$$y + 2 = 0 \Rightarrow y = -\frac{1}{2} \Rightarrow y = -\frac{1}{2}$$

$$T = Km^x L^y g^z$$

$$T = Km^0 L^{1/2} g^{-1/2}$$

$$T = K\sqrt{L \times \frac{1}{g}} \Rightarrow T = K\sqrt{\frac{L}{g}}$$

Mastered Dimension sheet Fundamental Quantities

mass [m]

Length [L]

Time [T]

Electric current [A]

Temperature [K]

Amount of substance [mol]

Luminous intensity [cd]

Kinematics

Area [L²]

Volume [L³]

Density [mL⁻³]

Velocity / speed [LT⁻¹]

Acceleration [LT⁻²]

Momentum [mLT⁻¹]

Dynamics

Force [mLT⁻²]

Impulse [mLT⁻¹]

Energy / work [mL²T⁻²]

Power [mL²T⁻³]

Pressure [mL⁻¹T⁻²]

Torque [m²T⁻²]

* How many cm² are there

In 1m²

$$1\text{m} = 100\text{cm}$$

$$1\text{m}^2 = (100)^2\text{cm}^2$$

$$1\text{m}^2 = 10^4\text{cm}^2$$

* Convert 90 km/h into m/s

$$90\text{ km/h} = \frac{90 \times 1000}{3600}$$

$$= 5 \times 5$$

$$= 25\text{ m/s}$$

* How many cm³ are there in 1m³

$$1\text{m} = 100\text{cm}$$

$$1\text{m}^3 = (100)^3\text{cm}^3$$

$$1\text{m}^3 = 10^6\text{cm}^3$$

ROTATIONAL & PERIODIC

Harshith

Frequency $[T^{-1}]$

Angular velocity $[T^{-1}]$

Angular momentum $[mL^2T^{-1}]$

Moment of Inertia $[mL^2]$

Properties of matter

Surface tension $[mT^{-2}]$

Viscosity (η) $[mL^{-1}T^{-1}]$

Elastic modulus $[mL^{-1}T^{-2}]$

Strain [Dimensionless]

Physical Constants

Gravitational constant $G [m^{-1}L^3T^{-2}]$

Planck's constant $[mL^2T^{-1}]$

Boltzmann Constant $k [mL^2T^{-2}K^{-1}]$

* The six limitations :-

Harshith

1) We can't find dimensionless constant.

$$\Rightarrow \text{Eg: } T = K\sqrt{\frac{L}{g}}, \quad v = K\sqrt{\frac{T}{m}}$$

2) Fail when there are more than Three unknowns.

3) Cannot distinguish between same dimension quantities.

$$\text{Eg: } [ML^2T^{-2}] \Rightarrow \text{work, Energy, Torque}$$

4) We cannot handle "+" (or) "-" in derivation

$$\text{Eg: } s = ut + \frac{1}{2}at^2$$

5) Can't handle trigonometry, logarithm, exponential

6) dimensionless correctness $\neq$ Physical correctness.

$$v = ut + 2at \rightarrow \text{Dimensionally correct}$$

$$v = ut + at \rightarrow \text{Physical correct?}$$

15/6/26

*

significant figures (SF) :- significant figures are the digits in a measurement that are meaningful and reliable.

Harshith

→ They tell you how precisely something has been measured.

more SF = more precision

*

SF Rules :-

i) Non-zero digits → Every non-zero digit always counts as significant.

Ex :- 234 → 3 SF

5923 → 4 SF

ii) Captive zeroes :- zeros sitting between two non-zero digits always count.

Ex :- 205 → 3 SF

7.024 → 4 SF

iii) Leading zeroes :- Zero before the first non-zero digit never count.

They only fix the decimal point.

Ex :- 0.005 → 1 SF

0.00042 → 2 SF

iv) Trailing zeroes (decimal) :- Trailing zeroes count where there is a decimal point.

Harshith

They show real precision.

$$\text{Ex: } 2.50 \rightarrow 3\text{SF}$$

$$\rightarrow 100.0 \rightarrow 4\text{SF}$$

v) Trailing zeros (no decimal) :- Ambiguous without a decimal point.

$\rightarrow$ scientific notation resolves it

$$*) 9 \times 10^3 \rightarrow 1\text{SF}$$

$$*) 3.0 \times 10^3 \rightarrow 2\text{SF}$$

$$*) 3.00 \times 10^3 \rightarrow 3\text{SF}$$

$$\text{Ex: } 2500$$

$$3.0 \times 10^3 \rightarrow 2\text{SF}$$

vi) Exact / counted numbers :- Counted or defined quantities are exact.

They have infinite significant figures

$$12 \text{ eggs} \rightarrow \infty \text{SF}$$

$$60 \text{ min/hr} \rightarrow \infty \text{SF}$$

★ Rounding Rules :-

Harshith

Next digit

< 5 (0, 1, 2, 3, 4)

Rule
Round Down
(Keep what you have)

Example

$4.734 \rightarrow 4.73$

≥ 5 (6, 7, 8, 9)

Round up (Add 1)

$4.736 \rightarrow 4.74$

Ex: - i) $2.3416 = 2.35$ ii) $7.635 \rightarrow 7.64$

iii) $4.892 = 4.89$

iv) $8.925 \rightarrow 8.92$

★ The Calculation Rules:

Addⁿ/subⁿ: Match "DECIMAL PLACES" of least precise Value.

mulⁿ/divⁿ: Match "SF"s" of least precise value.

Ex: - $3.52 + 2.3 + 1.421 = 7.241 = 7.2$

least decimal places = 1

Ans = 7.2

multiply

Harshith

$$\text{Eg: } 4.56 \times 1.4 = 6.384$$

Fewest SF = 2

$$\text{Ans} = 6.4$$

16/6/24

★ Errors:-

Error = Uncertainty in measurement

Error does NOT mean mistake.

Every measurement = True Value $\pm$ some uncertainty

★ Types of Errors:-

① Systemic Error:-

→ Def:- Always in the same direction.

• Always too high or always too low.

→ Cause:- Faulty or uncalibrated instrument.

→ Fix:- • Need a better instrument.

• Not more readings.

② Random error:-

→ Def: Unpredictable, sometimes high, sometimes low.

→ Cause: Vibrations, reaction time, fluctuations.

→ Fix: Take more readings → Average.

③ Gross Error:

→ Def: Oversight mistakes

→ Cause: misreading, wrong entry

→ Fix: Care and cross checking

more Readings → fixes RANDOM error only

more Readings → does NOTHING for SYSTEMIC Errors.

★ The Mean (Best Estimate)

Since the true value is unknown, we use

$$\bar{a} = \frac{a_1 + a_2 + a_3 + \dots + a_n}{n}$$

$\bar{a}$ is the best estimate of the true value.

★ Absolute Error:-

The deviation of each reading from the mean.

Always POSITIVE → take the magnitude.

$$\Delta a_i = |\bar{a} - a_i|$$

★ mean Absolute Errors

Harshith

$$\Delta \bar{a} = \frac{\Delta a_1 + \Delta a_2 + \dots + \Delta a_n}{n}$$

Eg :- Wire Diameter is measured 5 times :

$$4.52, 4.54, 4.51, 4.53, 4.52$$

$$a_1 \quad a_2 \quad a_3 \quad a_4 \quad a_5$$

$$= \frac{22.62}{5} = 4.524$$

$$4.524 = 4.52$$

$$\boxed{\bar{a} = 4.52}$$

$$\Delta a_i = |\bar{a} - a_i|$$

$$\Delta a_1 = |\bar{a} - a_1| = |4.52 - 4.52| = 0.00$$

$$\Delta a_2 = |\bar{a} - a_2| = |4.52 - 4.54| = 0.02$$

$$\Delta a_3 = |\bar{a} - a_3| = |4.52 - 4.51| = 0.01$$

$$\Delta a_4 = |\bar{a} - a_4| = |4.52 - 4.53| = 0.01$$

$$\Delta a_5 = |\bar{a} - a_5| = |4.52 - 4.52| = 0.00$$

$$\Delta \bar{a} = \frac{\Delta a_1 + \Delta a_2 + \Delta a_3 + \Delta a_4 + \Delta a_5}{5}$$

$$= \frac{0 + 0.02 + 0.01 + 0.01 + 0}{5}$$

$$= \frac{0.04}{5} = 0.008 = 0.01$$

Harshith

$$\boxed{\Delta \bar{a} = 0.01}$$

$$\text{Diameter} = \bar{a} \pm \Delta \bar{a} = 4.52 \pm 0.01$$

$$\begin{aligned} \star \text{Relative Error} &= \frac{\Delta \bar{a}}{\bar{a}} \\ &= \frac{0.01}{4.52} = 0.0022 \end{aligned}$$

$$\begin{aligned} \star \text{Percentage Error} &= \frac{\Delta \bar{a}}{\bar{a}} \times 100 \\ &= 0.0022 \times 100 = 0.22 \end{aligned}$$

Home work :- Length of Pendulum Readings (cm):

99.8, 100.2, 100.0, 99.6, 100.4

$$\begin{aligned} \text{Mean } \bar{a} &= \frac{99.8 + 100.2 + 100.0 + 99.6 + 100.4}{5} \\ &= \frac{500}{5} = 100 \end{aligned}$$

$$\Delta a_1 = |100 - 99.8| = 0.2$$

$$\Delta a_2 = |100 - 100.2| = 0.2$$

$$\Delta a_3 = |100 - 100.0| = 0$$

$$\Delta a_4 = |100 - 99.6| = 0.4$$

$$\Delta a_5 = |100 - 100.4| = 0.4$$

$$\Delta \bar{a} = \frac{1.2}{5} = 0.24$$

$$\text{Length} = 100 \pm 0.24 \text{ cm}$$

Realo

$$\text{Error} = 100 \pm 0.24$$

$$\text{Relative error} = \frac{\Delta a}{a} = \frac{0.24}{5} = 0.048$$

$$\text{Percentage error} = \frac{\Delta a}{a} \times 100 = 0.48\%$$

★ Propagation of Errors :-

When you calculate a result from measured quantities, their errors combine.

Key Principles :-

Errors always add, they never subtract.

Rule 1 :- Addⁿ / subⁿ : When two quantities are added or subtracted, their absolute errors add.

$$Z = A + B \Rightarrow \Delta Z = \Delta A + \Delta B$$

$$Z = A - B \Rightarrow \Delta Z = \Delta A + \Delta B$$

Eg :- Series Resistance

$$R_1 = [24 \pm 0.5] \Omega$$

$$R_2 = [8 \pm 0.3] \Omega$$

$$R = R_1 + R_2 = 24 + 8$$

$$\boxed{R = 32 \Omega}$$

$$\Delta R = \Delta R_1 + \Delta R_2$$

$$= 0.5 + 0.3$$

$$\boxed{\Delta R = 0.8}$$

$$\text{Result } (R \pm \Delta R) = (32 \pm 0.8) \Omega$$

The subⁿ warning :-

Harshith

$$A = (5 \pm 0.2) \quad B = (3.0 \pm 0.1)$$

$$Z = A + B = (8 \pm 0.3) \Rightarrow \frac{0.3}{8} = 0.0375 = 3.75\%$$

$$Z = A - B = (2 \pm 0.3) \Rightarrow \frac{0.3}{2} = 0.15 = 15\%$$

*) Rule No 2 :- mulⁿ / Divⁿ :

→ When two quantities are multiplied or divided, their relative errors add.

$$Z = A \times B \Rightarrow \frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

$$Z = A / B \Rightarrow \frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

Addⁿ / subⁿ :- Add Absolute Errors

mulⁿ / Divⁿ :- Add relative Errors

Eg :- Finding Resistance using Ohm's Law ($R = \frac{V}{I}$)

$$V = (10 \pm 0.5) \text{ V} \Rightarrow \frac{\Delta V}{V} = \frac{0.5}{10} = 0.05$$

$$I = (2 \pm 0.1) \text{ A} \Rightarrow \frac{\Delta I}{I} = \frac{0.1}{2} = 0.05$$

$$\frac{\Delta R}{R} = \frac{\Delta V}{V} + \frac{\Delta I}{I} = 0.05 + 0.05 = 0.10$$

Harshith

$$\frac{\Delta R}{R} = 0.10 \Rightarrow \Delta R = (0.1) R$$

$$R = \frac{V}{I}$$

$$R = \frac{10}{2} = 5$$

$$\boxed{R = 5} \rightarrow \Delta = (0.1) 5 = 0.5$$

$$\boxed{\Delta R = 0.5}$$

$$\text{Result: } (R \pm \Delta R) = (5 \pm 0.5)$$

Eg: - Area of a Rectangle : $L = (4 \pm 0.2) \text{ m}$

$$B = (3 \pm 0.3) \text{ m}$$

$$\frac{\Delta L}{L} = \frac{0.2}{4} \times 100 = \frac{20}{4} = 5\%$$

$$\frac{\Delta B}{B} = \frac{0.3}{3} \times 100 = \frac{30}{3} = 10\%$$

$$\frac{\Delta A}{A} = 0.15$$

$$\Delta A = 0.15 \times A$$

$$A = 4 \times 3 = 12$$

$$\boxed{A = 12}$$

$$\Delta = 0.15 \times 12$$

$$\Delta = 1.8$$

$$\text{Result: Area} = (12 \pm 1.8)$$

19/6/25

Rule No 3:- Power Rule:-

Harshith

→ When a quantity is raised by a power, its relative error is multiplied by that power.

$$Z = A^n \Rightarrow \frac{\Delta Z}{Z} = n \times \frac{\Delta A}{A}$$

General Formula:- $Z = \frac{A^p B^q}{C^r}$

$$\frac{\Delta Z}{Z} = p \left(\frac{\Delta A}{A} \right) + q \left(\frac{\Delta B}{B} \right) + r \left(\frac{\Delta C}{C} \right)$$

Eg:- Volume of a sphere $(V = \frac{4}{3} \pi r^3)$ is $r = (2 \pm 0.1) \text{ cm}$

$$\frac{\Delta V}{V} = 3 \times \frac{\Delta r}{r} = 3 \times \frac{0.1}{2}$$

$$= 3 \times 0.05$$

$$\frac{\Delta V}{V} = 0.15 \Rightarrow \Delta V = 0.15V$$

$$V = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi (2)^3 = 33.75 \text{ cm}^3$$

$$\Delta V = 0.15 \times 33.75 = 5 \text{ cm}^3$$

Result : $(V \pm \Delta V) = (33.75 \pm 5) \text{ cm}^3$

Ex - ② Pendulum : $g = \frac{4\pi^2 L}{T^2}$

Harshith

$$L = (100 \pm 0.5) \text{ cm} \quad T = (2 \pm 0.01) \text{ s}$$

$$\frac{\Delta g}{g} = 1 \times \frac{\Delta L}{L} + 2 \times \frac{\Delta T}{T}$$

$$= 1 \times \frac{0.5}{100} + \frac{0.01}{2} \times 2$$

$$= 0.005 + 0.01$$

$$= 0.015$$

$$\frac{\Delta g}{g} = 0.015$$

$$\Delta g = 0.015 \times g$$

$$g = \frac{4\pi^2 L}{T^2} \rightarrow \frac{4\pi^2 (100)}{2^2}$$

$$g = 9.8596 \times 100$$

$$\boxed{g = 985.96}$$

$$\text{Result} = (g \pm \Delta g) = (985.96 \pm 0.015)$$

(3)

Compound: $z = \frac{(a-b)}{(a+b)}$

$\frac{1.5 \pi \mu}{T} = \frac{e}{T}$ Harshith

Step 1:- Handle the ~~denominator~~ numerator

$$(a-b) = (0.5 \pm 0.2) = N$$

Step 2:- Handle the denominator

$$(a+b) = (3.5 \pm 0.2) = D$$

$$\frac{\Delta z}{z} = \frac{\Delta N}{N} + \frac{\Delta D}{D}$$

$$= \frac{0.2}{0.5} + \frac{0.2}{3.5}$$

$$= 0.4 + 0.0577$$

$$\frac{\Delta z}{z} = 0.457, \quad z = \frac{0.5}{3.5} = \frac{1}{7}$$

Q Density of a cylinder $\rho = \frac{4m}{\pi d^2 h}$
↓
Rho

$$m = (2 \pm 0.01) \text{ g}$$

$$d = (1 \pm 0.01) \text{ cm}$$

$$h = (4 \pm 0.01) \text{ cm}$$

$$\frac{\Delta \rho}{\rho} = 1 \times \frac{\Delta m}{m} + 2 \times \frac{\Delta d}{d} + 1 \times \frac{\Delta h}{h}$$

$$= \frac{0.01}{2} + 2 \times \frac{0.01}{1} + 1 \times \frac{0.01}{4}$$

$$= 0.005 + 0.02 + 0.0025$$

$$= 0.0075 + 0.02$$

$$\frac{\Delta \rho}{\rho} = 0.0275$$

Let $\rho = \text{mass} / \text{volume}$

Let $\rho = \text{mass} / \text{volume}$

Let $\rho = \text{mass} / \text{volume}$

Let $\rho = \text{mass} / \text{volume}$

Let $\rho = \text{mass} / \text{volume}$

Let $\rho = \text{mass} / \text{volume}$

20/6/20

Harshith

* Least count :- The smallest value an instrument can measure reliably.

Least count (LC)

Instrument

LC

→ 15 cm Rules / scale

$$1 \text{ mm} = 0.1 \text{ cm}$$

→ stop watch

$$0.015$$

→ Vernier Calipers

$$0.1 \text{ mm}$$

→ screw Gauge

$$0.01 \text{ mm}$$

Less LC = more precision.

MSD = Main scale division.

VSD = Vernier scale division.

$$10 \text{ VSD} = 9 \text{ MSD}$$

$$1 \text{ MSD} = 1 \text{ mm}$$

$$10 \text{ VSD} = 9 \times 1 \text{ mm}$$

$$1 \text{ VSD} = \frac{9}{10} \text{ mm}$$

$$1 \text{ VSD} = 0.9 \text{ mm}$$

$$LC = 1\text{msd} - 1\text{VSD} = 1\text{mm} - 0.9\text{mm}$$

$$LC = 0.1\text{mm}$$

* How to read a vernier scale:-

Step 1:- Read MSR (main scale Reading)

Step 2:- Note down coinciding no. of VS and MS.

Step 3:- Get VSR (vernier scale Reading) = Coinciding no. $\times$ LC.

Step 4:- Reading = MSR + VSR.

* Zero Error:-

Positive ^{zero} Error:- Right of main scale zero.

Negative zero Error:- Left of main scale zero.

$$\text{Correct Reading} = \text{Observed Reading} - \text{Zero Error}$$

Eg: observed = 1.24 cm

zero Error = 0.03

CR = 1.24 - 0.03 = 1.21 cm

Eg:- observed = 1.24 cm

Harshith

Zero error = -0.02 cm

$$\therefore CR = 1.24 \text{ cm} - (-0.02 \text{ cm})$$

$$= 1.26 \text{ cm.}$$

Eg: 20VSD coincide with 19msd

If 1 msd = 1 mm. Find the LC.

$$20 \text{VSD} = 19 \text{msd}$$

$$20 \text{VSD} = 19 \text{ mm}$$

$$1 \text{VSD} = \frac{19}{20} \text{ mm} = 0.95$$

$$1 \text{VSD} = 0.95$$

$$LC = 1 \text{msd} - 1 \text{VSD} = 1 \text{ mm} - 0.95 \text{ mm} = 0.05 \text{ cm}$$

Eg: main scale Reading MSR = 2.1, 5th vernier division coincide, least count 0.01

Harshith

$$2.1 \Rightarrow \text{MSR}$$

$$5 \Rightarrow \text{coincide}$$

$$0.1 \Rightarrow \text{LC}$$

$$5 \times 0.01 \rightarrow 0.05$$

$$2.1 + 0.05 = 2.15$$

Eg: MSR = 3.4 cm, 8th division coincides, LS is 0.01

$$3.4 \rightarrow \text{MSR}$$

$$8 \Rightarrow \text{coincide}$$

$$8 \times 0.01 = 0.08$$

$$0.08 \Rightarrow \text{VSR}$$

$$3.4 + 0.08 = 3.48 \text{ cm}$$

Eg: observed reading = 1.45, zero error = 0.03

$$1.45 - 0.03 = 1.42$$

Eg: observed reading = 1.45, zero error = -0.07

$$1.45 - (-0.07) = 1.45 + 0.07 = 1.52$$

Q.) In a vernier callipers, 50 vernier scale divisions are equal to 48 main scale divisions. If one main scale division = 0.05 mm, then the least count of the vernier callipers is _____ mm.

Sol: $50 \text{ VSD} = 48 \text{ MSD}$

$$1 \text{ MSD} = 0.05$$

$$1 \text{ VSD} = \frac{48}{50} \text{ MSD} = \frac{48}{50} \times 0.05$$

$$= \frac{48}{50} \times \frac{5}{100}$$

$$= \frac{48}{1000} = 0.048$$

$$1 \text{ VSD} = 0.048 \text{ mm}$$

$$\text{LC} = 1 \text{ MSD} - 1 \text{ VSD} = 0.050 - 0.048 = 0.002$$

$$\boxed{\text{LC} = 0.002 \text{ mm}}$$

★) Finding the value of zero Error:-

Harshith

step 1:- close the jaws

step 2 :- Find which vernier division coincides with a main scale line.

For Positive :-

$$\text{zero Error} = + (\text{Coinciding division}) \times LC$$

For negative :

$$\text{Zero Error} = -(n - \text{Coinciding division}) \times LC$$

$n = \text{Total no of vernier scale divisions.}$

$$n = 10.$$

$$LC = 0.1 \text{ mm} / 0.01 \text{ cm}$$

★) When both jaws of a vernier callipers touch each other, the zero mark the vernier scale is to the RIGHT of the zero mark of the main scale, and the 4th mark on the vernier scale coincides with a certain mark on the main scale, while measuring the length of the ~~length~~ of cylinder, the observed reads 15 divisions on the main scale and 5th division of the vernier scale coincides with a main scale division. The measured length of the cylinder is

— mm. (Least count of vernier calliper = 0.1 mm)

Harshith

Sol: $+VE \quad ZE = 4 \times LC = 4 \times 0.1 = 0.4 \text{ mm}$

$$MSR = 15 \times 1 \text{ mm} = 15 \text{ mm}$$

$$VSR = 5 \times 0.1 \text{ mm} = 0.5 \text{ mm}$$

$$\text{Observed Reading} = MSR + VSR = 15.5 \text{ mm}$$

$$CR = OR - ZE = 15.5 - 0.4 = 15.1 \text{ mm}$$

Q3) With the jaws closed the 8th division (of 10) coincides and the vernier zero is to the left of the main-scale zero. While measuring an object the main scale reads 2.0 cm and the 5th vernier division coincides. (Least count = 0.01 cm). Find the corrected length.

Sol: $-VE \quad ZE = -(10 - 8) \times 0.1 \text{ mm}$

$$-VE \quad ZE = -0.2 \text{ mm}$$

$$MSR = 20 \text{ mm}, \quad VSR = 5 \times 0.1 \text{ mm} = 0.5 \text{ mm}$$

$$OR = MSR + VSR = 20.5 \text{ mm}$$

Harshith

$$CR = OR - ZE = 20.5 - (-0.2) = 20.7 \text{ mm} / 2.07 \text{ m.}$$

23/6/26

★) Screw Gauge

*} Pitch:- Screw Gauge works like a screw. When you turn the screw one full round, it moves forward by a fixed distance.

That fixed distance is called as "Pitch".

→ standard values

$$\text{Pitch} = 1 \text{ mm} / 0.5 \text{ mm}$$

★ Least count

$$LC = \frac{\text{Pitch}}{\text{no. of circular divisions}}$$

Eg: $P = 1 \text{ mm}$, $CSD = 100$, LC ?

$$LC = \frac{P}{CSD} = \frac{1}{100} = 0.01 \text{ mm.}$$

$$\text{Eg: } p = 0.5 \text{ mm}, \text{CSO} = 50, \text{LC} = ?$$

Harshith

$$\text{LC} = \frac{0.5}{50} = 0.01$$

★) How to read a reading on SG.

$$\text{Total Reading} = \text{PSR} + (\text{CSR} \times \text{LC})$$

PSR = Pitch scale Reading, (last fully visible mark on the main scale).

CSR = Circular scale Reading. (Which circular division lines up with reference line)

$$\text{Eg: } P = 1 \text{ mm}, 100 \text{ division}, \text{LC} = 0.01 \text{ mm}$$

$$\text{PSR} = 5 \text{ mm}, \text{CSR} = 28$$

$$5 + (28 \times 0.01)$$

$$= 5 + 0.28$$

$$= 5.28 \text{ mm}$$

Eg: PSR = 3

CSR = 65

Harshith

$$= 3 + (65 \times 0.01)$$

$$= 3 + 0.65$$

$$= 3.65$$

* Zero Error :-

Step 1 :- Close the screw fully

Step 2 :- Look at where the circular scales 0 line sits relative to the reference line on the scale.

* Positive Zero Error :-

Def:- Circular scales zero BELOW the Reference line. This means extra circular divisions are showing above the reference line.

Instrument reads too high

$$\text{Positive ZE} = \text{Coinciding circular division} \times \text{LC}$$

* Negative zero error!

Harshith

Circular scales zero ABOVE the reference line.
This means the circular scale has not rotated far enough to reach zero.

Instrument reads too low.

$$\text{Negative ZE} = -(n - \text{coinciding CD}) \times LC$$

$$n = \text{Total no. of circular}$$

$$\text{Correct Reading} = \text{Observed Reading} - \text{ZE}$$

24/6/26

* Problems :-

Q1) 100 divisions on a circular scale. The 3rd division lies on the reference line & is below the reference line.
Find zero error.

Sol: ZERO Below $\Rightarrow$ +VE ZE

$$+VE ZE = \text{coinciding circular division} \times LC$$

$$\Rightarrow 3 \times 0.01 = 0.03 \text{ mm}$$

$$\Rightarrow 0.03 \text{ mm}$$

Q2) 100 divisions on circular scale, the 97th division lies on the reference line. 0 is above the reference line. Find zero error.

Sol: Zero Above $\Rightarrow$ -VE ZE

$$\therefore -(n - \text{Coinciding CD}) \times LC$$

$$-(100 - 97) \times 0.01 = -0.03 \text{ mm}$$

Q3) 50 divisions on a circular scale, the 5th division lies on the reference line. Zero is below the reference line. Find zero error.

Sol: +VE ZE = $5 \times 0.01 = 0.05 \text{ mm}$

Q4) 50 divisions on a circular scale, the 44th division lies on the reference line. Zero is above the reference line. Find zero error.

Sol: $-(50 - 44) \times 0.01 \text{ mm}$

$$-(6) \times 0.01 \text{ mm} \Rightarrow -0.06 \text{ mm}$$

Q) Pitch = 1mm, circular scale = 100 divisions, when screw is closed 4th circular division is on the Reference line, zero is below. We are measuring a wire whose, $PSR = 2mm$ and $CSR = 56$ find the corrected diameter.

Sol: $L_c = \frac{\text{Pitch}}{CS} = \frac{1mm}{100} = 0.01mm$

Zero Below $\Rightarrow +ve ZE$

$$+ve ZE = +4 \times 0.01 = 0.04mm$$

$$\boxed{OR = PSR + (CSR \times LC)}$$

$$OR = 2mm + (56 \times 0.01) = 2.56mm$$

$$CR = OR - ZE$$

$$CR = 2.56 - 0.04$$

$$CR = 2.52mm //$$

Q1) $P = 1\text{mm}$, Circular scale = 100 divisions, when screw is closed the 96th circular division is on the reference line. Zero is above.

We are measuring a thin sheet whose

$PSR = 1\text{mm}$, $CSR = 30$. Find the corrected thickness.

Sol:
$$L_c = \frac{\text{Pitch}}{CD} = \frac{1}{100} = 0.01\text{mm}$$

Zero Above $\Rightarrow$ -ve ZE

$$\text{-ve ZE} = -(n - \text{Coinciding CD} \times L_c)$$

$$= -(100 - 96 \times 0.01)$$

$$= -4$$

$$= -(4 \times 0.01) = -0.04\text{mm}$$

$$\text{-ve ZE} = -0.04\text{mm}$$

$$OR = PSR + (CSR \times L_c)$$

$$= 1\text{mm} + (30 \times 0.01) = 1.30\text{mm}$$

$$CR = OR - ZE$$

$$= 1.30 - (-0.04) = 1.34\text{mm}$$